

Neurons as EMF Propagation Structures — Signal Geometry, Chronic Pain, and Fibromyalgia as Global Field-Mapping Failure

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Abstract

Neurons are electromagnetic field (EMF) propagation structures. They are not chemical relay stations that happen to produce electrical side effects. The action potential is the forward edge of a collapsing and regenerating electromagnetic field moving through a living, geometrically complex, three-dimensional medium. Myelin, nodal spacing, axon diameter, dendritic branching angles, and the dielectric properties of surrounding glial and extracellular tissue all determine the shape, velocity, and spatial extent of that propagating field. Signal geometry — the three-dimensional trajectory, amplitude envelope, and temporal pattern of the wavefront — is not incidental to neural signaling. It is the signal.

This paper argues that chronic pain is not primarily a disease of tissue chemistry. It is a signal geometry failure. The nervous system is propagating a legitimate EMF disturbance along a pathway whose geometry has been altered — by inflammation, tissue remodeling, demyelination, or altered nodal spacing — and is interpreting the distorted signal as ongoing harm because the distorted field arrives at the wrong cortical address, at the wrong time, with the wrong spatial profile. The brain does the only thing it can: it reads the geometry it receives, and misread geometry produces mislocated, misdated, or amplified pain.

Fibromyalgia is the systemic version of this failure. It is not a peripheral pathology; it is a global collapse of field-mapping coherence across the entire nociceptive network. Central sensitization at scale, disrupted descending inhibitory field geometry, thalamocortical dysrhythmia, small fiber neuropathy, chronic glial activation, and the nightly failure of slow-

wave sleep architecture converge to produce a nervous system that cannot correctly map any signal it receives. The result is bodywide pain without a bodywide injury.

Current diagnostic and therapeutic models treat pain as a chemical message problem. This paper argues it is a geometry and field-propagation problem — and that distinction explains why chemical interventions fail so reliably for so many patients, and what a geometry-first treatment strategy would actually look like.

1. Introduction — The Wrong Mental Model

Medicine inherited a telephone-wire model of the neuron. A signal originates at the periphery, travels down the axon to the terminal, chemicals are released, the next neuron fires, and so on up to the brain. The message arrives, pain is felt, the hand withdraws. Clean, linear, intuitive. And for reflex pathways and simple acute signaling, it is good enough. It breaks down completely the moment the clinical question gets more complicated.

Phantom limb pain cannot be explained by the telephone model, because the wire has been cut. The limb is gone, the nerve is severed, and yet the pain continues — often precisely in the missing hand or foot. Referred pain cannot be explained by the telephone model, because the signal is arriving from a location that has nothing to do with where the patient reports the pain. Fibromyalgia cannot be explained by the telephone model, because there is no consistent tissue damage, no consistent inflammatory lesion, no single broken wire. Central sensitization — where the nervous system amplifies pain in the complete absence of any ongoing peripheral injury — is, by the telephone model's logic, simply impossible. The model says: pain requires injury, injury sends a signal, the signal produces pain. No injury, no signal. No signal, no pain.

Every one of those clinical realities is happening in millions of patients right now. So the model is wrong. Not slightly wrong, not approximately wrong — structurally wrong, wrong at the level of its core metaphor.

The telephone model asks the wrong question. It asks: what chemical is elevated or depleted, and how do we adjust it? The geometry model asks: what is the shape of the field doing, where is it arriving, and does that arrival pattern match the brain's existing map? These are different questions. They lead to different experiments, different diagnostics, and radically different treatments.

The core reframe of this paper is this: neurons do not transmit messages the way copper wire transmits current. They propagate electromagnetic field disturbances through a living, changing, three-dimensional medium. The signal is the field. The field has shape. That shape is its meaning. Alter the shape — through any mechanism, chemical, mechanical, structural, or inflammatory — and you alter the meaning the brain extracts from it. In the pain system, a meaning error produces pain that does not match reality.

Section 3 establishes the biophysics: what it means to call a neuron an EMF propagation structure. Section 4 introduces signal geometry as the framework: how the shape of a propagating wavefront encodes location and quality information. Section 5 applies this framework to chronic pain: what breaks, how it breaks, and what the clinical syndromes look like through a geometry lens. Section 6 recasts the Gate Control Theory of Melzack and Wall within a field geometry framework. Section 7 applies the full framework to fibromyalgia, arguing it represents a system-level geometry collapse rather than any single pathological process. Section 8 examines why current diagnostic methods are poorly suited to detecting geometry failures. Section 9 translates the geometry framework into practical therapeutic directions. Section 10 draws conclusions and calls for a redirection of pain research toward geometry as the primary variable.

2. Neurons as EMF Propagation Structures

The action potential is taught as a voltage event. Sodium ions rush in, the membrane depolarizes, potassium ions flow out, the membrane repolarizes, and the whole thing happens again a millimeter down the axon. This description is accurate at the level of ion flux. It is incomplete at the level of physics.

Every moving charge separation produces an electromagnetic field. The action potential is not just a voltage spike at the membrane — it is the forward edge of a propagating electromagnetic disturbance. As the depolarization wave advances along the axon, it creates a moving electric dipole moment: a zone of net positive charge ahead of the wave front, a zone of net negative charge behind it. That dipole radiates. The field extends into the extracellular space, into neighboring axons, into the surrounding glial matrix, and into the interstitial fluid. It is not contained within the axon membrane. The axon is not a wire. It is a field emitter moving through a conducting and dielectric medium.

This was implicit in the Hodgkin-Huxley model of 1952, which mathematically described the ionic basis of the action potential with precision that earned it the Nobel Prize. The model is framed in terms of membrane conductance and ion channel kinetics. It does not ignore the electromagnetic nature of the event — the mathematics of conductance and capacitance are electromagnetic mathematics — but the field interpretation was not the framing Hodgkin and Huxley chose, and it is not the framing medical education has inherited. The result is that generations of clinicians have been trained to think in terms of chemicals and voltages rather than fields and geometries.

Myelin changes the geometry of the field, not merely its speed. In a myelinated axon, the membrane is insulated between nodes of Ranvier. The depolarizing current cannot flow through the insulated internode — so it flows longitudinally inside the axon and jumps to the next node, where it triggers a new action potential. This is saltatory conduction. The important point for this paper is that each node of Ranvier is a field regeneration point.

The propagating electromagnetic disturbance does not degrade and amplify continuously — it collapses and resets at discrete spatial intervals. The spacing of nodes sets the spatial frequency of the signal. Alter the spacing — through demyelination, edema, or the microstructural changes that accompany chronic inflammation — and you change the spatial frequency of the propagating field. The signal that arrives at the next synapse, the next relay nucleus, is not the same signal that left the periphery. Its geometry has been corrupted in transit.

Axon diameter determines propagation velocity and the spatial extent of the field gradient. The largest myelinated axons — A-alpha fibers, governing proprioception and motor control — propagate at 70 to 120 meters per second and generate broad, rapidly advancing field disturbances. The smallest unmyelinated C fibers — carrying slow, burning, chronic pain — propagate at 0.5 to 2 meters per second. They generate narrow, slow-moving fields. These are not just speed differences. They are geometric differences in the field disturbance that arrives at the dorsal horn and eventually at the cortex. The brain reads different geometry from A-alpha and C-fiber input not just because of which receptor initiated the signal, but because the field shape of the arriving wavefront is categorically different.

Dendritic trees are not passive receivers. A neuron with a highly branched dendritic arbor — the pyramidal cells of the cortex, the Purkinje cells of the cerebellum, the wide-field neurons of the dorsal horn — is a three-dimensional antenna array. The branching geometry determines which field orientations and spatial frequencies the neuron is most sensitive to. A field arriving along the somatodendritic axis produces a different pattern of membrane polarization than the same-amplitude field arriving orthogonally. The neuron does not merely sum inputs; it performs a geometric computation on the spatial structure of the fields reaching it. This is not metaphor — it follows directly from the biophysics of cable theory extended to three-dimensional morphologies, as Anastassiou and colleagues have formally demonstrated for cortical neurons subject to extracellular field fluctuations.

Local field potentials — the summed electromagnetic activity of many neurons firing in a volume of tissue — are not noise or a byproduct. They are the medium through which

individual axons propagate. Neurons are not isolated wires running through neutral space. They are embedded in a collective field generated by their own coordinated activity. Anastassiou et al. demonstrated in 2011 that this collective field feeds back onto individual neurons through ephaptic coupling — direct field-to-membrane interaction without chemical synaptic transmission. The geometry of the collective field shapes the firing probability of every neuron within its spatial extent. When the local field potential changes its oscillatory frequency, its amplitude, or its spatial distribution, every neuron in that volume changes its response characteristics. This is a tissue-level electromagnetic architecture, not a collection of independent signal cables.

The astrocytes, microglia, oligodendrocytes, and the extracellular matrix all participate in defining the dielectric environment through which neural fields propagate. Astrocytes regulate extracellular potassium concentration, which directly affects the electrochemical gradient that drives ion flow. They couple electrically through gap junctions, forming a syncytium that can redistribute field-relevant charge across millimeters of tissue. The interstitial fluid's ionic composition sets the conductivity of the extracellular medium. Change the conductivity — through inflammation, edema, or neuroinflammatory activation — and you change the propagation characteristics of every field in that tissue volume. These are not peripheral details. They are the substrate of signal geometry.

3. Signal Geometry — How Shape Determines Meaning

In the nervous system, where a signal goes is part of what it means. This is not a philosophical observation. It is a physical fact about how the somatosensory system is organized. The primary somatosensory cortex maintains a precise topographic map of the body — the homunculus. Specific regions of cortex receive input from specific body regions. The cortical area representing the hand is adjacent to the cortical area representing the face, not the foot. This topographic organization is the nervous system's spatial labeling

system. There is no separate tag on each neural signal that says "this came from the left thumb." The spatial address of arrival in the cortex is the label.

Pain localization, therefore, is a geometric achievement. A signal that correctly propagates from a nociceptor in the left thumb, through the correct dorsal horn segment, up the correct spinothalamic tract column, through the correct thalamic relay nucleus, and to the correct patch of somatosensory cortex — that signal produces correctly localized pain: sharp, left thumb, now. Every step in that propagation is a geometry step. The axon must be the right diameter, the myelin must be intact, the nodal spacing must be correct, the fascicle orientation must be preserved, the thalamic relay must be firing at the right oscillatory frequency to gate the input to the right cortical column. Geometry all the way through.

Temporal geometry matters as much as spatial geometry. The precise timing of arrival at a synapse determines whether summation occurs. If two inputs arrive within the right temporal window — a few milliseconds for fast synapses — they sum and cross threshold together, producing a postsynaptic action potential. Arrive outside that window, and there is no summation, no action potential, no signal propagation. A signal that is 2 milliseconds late relative to its expected convergence partner may fail to produce pain — or may produce a different quality of pain, or an undifferentiated background noise that the brain cannot resolve into a specific sensation. Temporal geometry is also the substrate of long-term potentiation and depression. Whether a synapse strengthens or weakens after stimulation depends on the precise temporal ordering of pre- and postsynaptic activity — spike-timing-dependent plasticity. The chronic pain field has not fully reckoned with the fact that this timing-dependence means that chronic pain is, in part, a learned temporal geometry pattern that has been consolidated into synaptic strength.

The nervous system does not maintain a separate catalog that associates each signal with a sensation label. The geometry is the catalog. The quality of pain — sharp versus burning versus aching versus throbbing — emerges from the fiber type, the spatial extent of activation, the temporal frequency of firing, and the pattern of cortical and limbic areas

recruited by the arriving field. A C-fiber signal arriving at low frequency produces a different quality of experience than the same C-fiber signal arriving at high frequency. A signal that recruits the anterior cingulate cortex (which contributes to the emotional component of pain) in addition to the somatosensory cortex produces a different experience than the same signal reaching only the somatosensory cortex. These qualitative differences are not stored in a molecule. They are in the geometry of which neural populations are activated, in what spatial arrangement, and in what temporal order.

This has a stark clinical implication: if you disrupt the geometry anywhere along that chain, you disrupt the meaning. And disrupted meaning in a pain-signaling system produces pain that does not correctly represent the state of the body. Not imaginary pain. Not psychosomatic pain. Real neural field activity arriving at real cortical addresses — but at the wrong addresses, or in the wrong temporal pattern, or with the wrong spatial envelope. The brain interprets what it receives as best it can, fitting the distorted input to its existing maps. Where the maps are close, the pain is close to accurate. Where the distortion is severe, the pain may be wildly inaccurate — referral, mislocation, amplification — but it is not invented. It is the brain correctly reading incorrect geometry.

4. Chronic Pain as Mis-Mapped Signal Geometry

Acute pain works. A hammer strikes the thumb. Nociceptors discharge. A-delta fibers carry the sharp first pain with correct geometry: fast, spatially precise, localized. C-fibers follow with the burning second pain: slower, more diffuse, but still geometrically constrained to the injured region. The field arrives at the correct dorsal horn segment, travels the correct spinothalamic column, reaches the correct thalamic relay, activates the correct cortical column, and the brain reads: left thumb, sharp, now, significant. Withdrawal reflex, learned avoidance, wound care — the entire adaptive pain response executes correctly. Geometry intact, function intact.

Chronic pain is what happens when the geometry breaks. The signal is still propagating. There may still be a genuine peripheral source — or there may not. But the pathway has been altered. Inflammation changes the ionic environment around peripheral nociceptors, sensitizing them and lowering their discharge threshold. The resulting barrage of high-frequency C-fiber input into the dorsal horn triggers changes in spinal neuron excitability that alter the spatial and temporal geometry of signal propagation at the central level. Tissue remodeling following injury changes the mechanical environment of nerves — compression, stretch, angulation — changing the spatial arrangement of axons within fascicles and altering field propagation geometry. Demyelination, whether from direct nerve injury, autoimmune processes, or nutritional deficiency, alters nodal spacing and node-of-Ranvier properties, corrupting the spatial frequency of the propagating field. The signal that arrives at the brain is not the signal that left the periphery. It has been deformed in transit.

The brain reads the deformed signal against its existing maps. The somatosensory cortex has a prior — an expectation of what signals from a given body region look like. When the arriving field matches that prior closely enough, the brain confidently localizes the pain. When the arriving field is distorted, the brain still tries to fit it to the closest map address. The result is pain in a location that does not perfectly correspond to any actual injury site. Pain that changes location. Pain that spreads beyond the injury zone. Pain that the patient describes with frustration: "It moves." The pain moves because the distorted field is matching different cortical addresses as its geometry shifts, not because there is a traveling injury.

Central sensitization, as characterized by Woolf in his definitive 2011 review, is the nervous system's geometry expanding in the wrong direction. Spinal dorsal horn neurons lower their activation threshold and expand their receptive fields — the zone of peripheral tissue from which they can be driven. This is a geometric change. A neuron that previously responded only to input from a 2-centimeter patch of skin now responds to input from a 10-centimeter zone. Any signal arriving from that wider zone now accesses the nociceptive

pathway. The spatial geometry of "what counts as pain input" has grown. The amplifier has been turned up, and the antenna has been widened. Both are geometry changes, not chemistry changes — or more precisely, they are chemistry changes whose functional consequence is a geometry change.

Windup is temporal geometry going wrong. When C-fibers are stimulated at repetitive intervals below about 0.3 Hz, dorsal horn neurons show progressively increasing responses — each stimulus produces a larger output than the last, even though the peripheral input is constant. This is not because more neurotransmitter is being released. It is because the temporal pattern of the field input is now summing in a way that keeps NMDA receptor channels open — they are "wound up." The geometry of the temporal signal — the interval between pulses — is the mechanism. Hit the same nerve with the same stimulus at longer intervals and no windup occurs. The time pattern is the key, which is to say, the temporal geometry is the mechanism.

Referred pain is a map collision. Afferent fibers from the heart and from the left arm converge on the same dorsal horn segments — primarily C7, T1, and T2. The spinal neurons at those segments receive input from both sources. When the heart sends an ischemic distress signal, the field arrives at the same dorsal horn geometry that normally receives left-arm input. The thalamus and cortex receive a signal that looks, geometrically, like left-arm input. The brain has no mechanism to identify that this particular instance of "left-arm geometry" came from the heart instead. So it localizes the pain to the left arm, shoulder, and jaw — because those are the addresses this geometry normally represents. This is not a failure of intelligence. It is the correct operation of a system that uses geometry as its labeling system, when two sources share a geometric address.

Phantom limb pain is the clearest demonstration that pain is a cortical geometry phenomenon. The amputation removes the peripheral structure. The dorsal horn segment, the thalamic relay, and the cortical column representing the lost limb remain physically intact. The cortical map has not been erased — it is still there, still capable of generating the experience of sensation in the missing hand. Input from adjacent body regions — the

residual limb, nearby trunk — invades the cortical territory formerly dedicated to the missing limb. That invasion follows the geometry of cortical adjacency: the face cortex is immediately adjacent to the hand cortex in the homunculus, which is why many amputees report that touch to the face evokes referred sensation in the phantom hand. The field still propagates into the old cortical address. The brain still reads it as coming from the missing limb, because that is what signals at that geometric address have always meant.

Allodynia — pain from light touch, from clothing, from a gentle breeze — is a threshold geometry failure. Under normal conditions, low-threshold mechanoreceptor input via A-beta fibers activates inhibitory interneurons in the dorsal horn that prevent that input from accessing the nociceptive pathway. The geometric separation between the touch pathway and the pain pathway is maintained by active inhibitory architecture. When central sensitization erases that inhibitory architecture — when the inhibitory interneurons are suppressed, when their geometry of influence collapses — A-beta input now geometrically accesses the nociceptive pathway. Light touch becomes pain, not because touch receptors have changed, but because the gateway geometry that kept them out of the pain pathway has been dismantled.

5. The Gate Control Theory Reframed — Geometry, Not Gates

Melzack and Wall published the Gate Control Theory in 1965 in *Science*, and it was the most important conceptual advance in pain neuroscience of the twentieth century. The theory proposed that pain transmission cells in the spinal dorsal horn are modulated by a gate: a balance between large-diameter A-beta fiber input (touch, pressure, vibration) and small-diameter C- and A-delta fiber input (pain). Large-fiber activation drives inhibitory interneurons that close the gate. Small-fiber activation inhibits those interneurons, opening the gate. The brain can also influence the gate from above through descending control pathways. Pain is not a fixed signal; it is a modulated one.

That conceptual framework has held up remarkably well. But the gate metaphor — while vivid and useful — has been interpreted primarily in chemical terms. Inhibitory interneurons release GABA and glycine. Block those inhibitory synapses, open the gate. Promote them, close the gate. Adjust the chemistry, adjust the pain. This chemical interpretation has driven pharmacological approaches that work some of the time in some patients and fail most of the time in chronic pain.

Here is the geometry reframe: the gate is not an inhibitory interneuron balancing act. The gate is a geometric filter. When large-diameter A-beta fibers are active, they propagate a fast, broad-field disturbance into the dorsal horn. That disturbance, through the ephaptic and synaptic mechanisms described in Section 3, activates inhibitory interneurons and alters the local field potential in the dorsal horn in a way that makes small-fiber C-fiber input geometrically unable to propagate past the first synapse to the projection neurons. The inhibitory field generated by A-beta activity physically reshapes the electromagnetic environment of the dorsal horn circuit. C-fiber signals arrive into that environment and cannot sum to threshold at the projection neuron. The gate is closed.

When A-beta fibers are damaged — by nerve compression, by the demyelination that often accompanies diabetes, by the small-fiber loss of aging — the geometric filter disappears. The dorsal horn's local field potential is no longer structured by the ongoing large-fiber input. C-fiber signals now propagate into an unfiltered electromagnetic environment. They reach projection neurons without interference. The gate stands open. Chronic pain follows not because a chemical has changed but because the geometric filter has been physically removed.

Transcutaneous Electrical Nerve Stimulation works because it re-imposes A-beta-frequency field disturbances from outside the body. The electrodes deliver electrical current at frequencies and amplitudes that activate large-diameter sensory fibers preferentially — because large fibers have lower electrical thresholds than small fibers. The externally generated field pattern mimics the endogenous A-beta activity that would normally generate the geometric filter in the dorsal horn. TENS is, in the geometry

framework, an external field geometry restoration device. It is not "distracting" the nervous system. It is artificially regenerating the inhibitory field structure that chronic pain or nerve damage has erased.

This reframing has a direct and uncomfortable therapeutic implication. If the physical geometry of the gate mechanism has been altered — if A-beta fibers have been damaged, if dorsal horn interneurons have been lost, if the tissue architecture that shapes large-fiber field propagation has been remodeled — you cannot fully restore the gate with chemistry. You can modulate individual synapses with GABAergic drugs. You can reduce C-fiber input with sodium channel blockers. But you cannot chemically reconstruct the physical geometry of the filter. To restore the gate, you need to restore the geometry. That requires physical, structural, or electrophysiological interventions — not only pharmacological ones.

6. Fibromyalgia — Global Field-Mapping Failure

Fibromyalgia has no consistent peripheral pathology. Decades of research have found no single tissue lesion, no uniform inflammatory marker, no structural abnormality that appears reliably across the patient population. This absence of peripheral pathology has made fibromyalgia one of the most controversial diagnoses in medicine. A substantial fraction of clinicians have historically suspected that the disease is psychosomatic — that the pain is generated by psychological distress rather than any biological process. This view is wrong, and the EMF geometry framework makes it clear why it is wrong, and what is actually happening instead.

Fibromyalgia is not a peripheral problem. It is a system-level failure of field-mapping coherence across the entire nociceptive network. Consider a GPS system where all the satellites are functioning, all the receivers are functioning, and all the signals are being transmitted correctly — but the coordinate system embedded in the receiver software has become internally inconsistent. Every signal arrives, every calculation runs, and every

output is wrong. Not dramatically wrong in every case — some position fixes are close to accurate — but wrong in a systematic, pervasive way that makes the system unreliable for navigation. That is fibromyalgia. The peripheral nociceptors are firing. The spinal cord is transmitting. The thalamus is relaying. The cortex is processing. But the coordinate system — the geometry that links peripheral events to cortical addresses — is incoherent throughout, and the brain perpetually cannot correctly map what it is receiving. In a pain-mapping system, failed mapping defaults to pain. The brain cannot resolve the incoherent field into a specific, localized sensation, so the output is diffuse, widespread, and continuous.

The first mechanism is central sensitization at scale. Woolf's framework describes central sensitization at the level of individual dorsal horn segments — specific neurons lowering their threshold in response to a specific peripheral input. In fibromyalgia, this process has occurred across the entire spinal cord. Multiple dorsal horn segments are simultaneously hyperexcitable. The thalamic relay nuclei are running hot. The somatosensory cortex shows elevated baseline activity. The global field sensitivity of the nociceptive network has been increased, not in one location, not in response to one injury, but system-wide. Every signal entering this system encounters an amplifier that has been turned to maximum gain with the contrast knob also turned up. Low-level peripheral input that would normally be filtered out now propagates all the way to cortical awareness as pain.

The second mechanism is disrupted descending inhibitory field geometry. The periaqueductal gray (PAG) and the rostral ventromedial medulla (RVM) are the primary sources of descending pain modulation. Under normal circumstances, these structures broadcast descending inhibitory signals that impose geometric order on ascending nociceptive traffic — essentially, they modulate the local field potential at dorsal horn levels in a way that suppresses nociceptive transmission. This is diffuse noxious inhibitory control (DNIC), also now called conditioned pain modulation (CPM). In fibromyalgia patients, CPM is consistently abnormal. The descending geometry is not being broadcast with normal strength or spatial coherence. The ascending nociceptive pathway is running

without its normal top-down geometric constraint. Signals that would have been suppressed by descending inhibition are not suppressed. They propagate upward and produce pain.

The third mechanism is thalamocortical dysrhythmia. The thalamus acts as a field synchronizer for cortical processing. It broadcasts rhythmic oscillatory activity — theta and alpha-band oscillations at rest, different patterns during active processing — that gates and coordinates cortical activity. In fibromyalgia, EEG and MEG studies have documented abnormal thalamocortical oscillatory patterns: excess alpha-band activity in regions that should show beta or gamma activity during sensory processing, disrupted synchrony between thalamic relay nuclei and cortical areas. The rhythm that normally gates and routes pain signals through the cortex is out of phase. Signals arrive at the cortex but they are not being correctly routed by the oscillatory gating mechanism. The thalamus is supposed to act like an air traffic controller, sequencing signals into their correct cortical landing zones. In fibromyalgia, the controller is calling out incorrect runways. Every landing is approximate, many are wrong, and the consequence is that the cortex is perpetually trying to process a traffic pattern that does not make sense.

The fourth mechanism is small fiber neuropathy. A significant and reproducible subset of fibromyalgia patients — estimates range from 40 to 60 percent depending on the study and methodology — have measurable intraepidermal nerve fiber density reduction on skin punch biopsy. This is structural peripheral pathology. The fine-caliber C-fibers and A-delta fibers that serve as the peripheral input to the nociceptive system are physically reduced in number. The remaining fibers show morphological changes consistent with neuropathic remodeling. This means the peripheral EMF input into the system is already distorted before it ever reaches the spinal cord. An already-corrupted peripheral field signal is being fed into an already-overwhelmed and mistuned central processor. The system is receiving bad input and processing it badly. The wonder is not that fibromyalgia patients are in pain — it is that their nervous systems continue to function at all in any organized way.

The fifth mechanism is glial activation. Astrocytes and microglia in fibromyalgia show chronic inflammatory activation — elevated cytokines, altered morphology, disrupted potassium buffering. As established in Section 3, these cells are not bystanders to neural signaling. They are part of the dielectric tissue matrix that shapes the EMF environment of local circuits. Chronically activated glia change the impedance landscape of the tissue — altering extracellular ionic concentrations, changing gap junction coupling, releasing neuroactive substances that modify neuronal excitability. The physical substrate through which neural fields propagate has been altered. The medium has changed, so the signals propagating through the medium have changed. Staud and colleagues have documented evidence of neuroinflammatory processes in fibromyalgia that support this mechanism, and Hains and Waxman established in animal models that spinal microglial activation is sufficient to produce chronic widespread pain sensitization through exactly the kind of dielectric tissue-property changes described here.

The sixth mechanism is the failure of sleep architecture as a nightly field reset. Deep slow-wave sleep — delta-wave sleep — is the period during which the brain runs its maintenance and recalibration protocols. Thalamocortical oscillations shift to slow, high-amplitude delta rhythms. The default mode network and somatosensory networks show reorganized activity patterns consistent with memory consolidation and sensory map maintenance. This is when the nervous system has the opportunity to reset the geometric ordering of its processing architecture. Fibromyalgia patients have well-documented disruption of slow-wave sleep — specifically, the intrusion of alpha-frequency oscillations into delta sleep, a phenomenon first described by Moldofsky in the 1970s and replicated many times since. The nightly field geometry reset is not happening. The system wakes each morning more mis-mapped than it was when it went to sleep. The geometry failure compounds nightly. Treating sleep is not a secondary or supportive intervention in fibromyalgia — it is foundational. Without restoring slow-wave sleep, every other intervention is fighting against a system that is being re-corrupted every night.

This multi-mechanism framework explains the fibromyalgia symptom profile with precision that no single-mechanism theory has achieved. Widespread pain exists because the whole map is corrupted — there is no single region where geometry is preserved. Fatigue exists because the nervous system is expending enormous metabolic energy attempting to resolve signals it cannot correctly interpret — the computational load of perpetual ambiguity is exhausting, literally and physiologically. Cognitive fog — "fibro fog" — exists because the same thalamocortical geometry that maps body sensations also gates attention, working memory, and executive function. The thalamus that cannot correctly route pain signals also cannot correctly route cognitive signals. Sleep disruption is upstream of the pain, not downstream — it is one of the primary drivers of the geometry failure, not a consequence of being in pain at night, though it becomes both over time. Sensitivity to temperature, pressure, and sensory overload exists because the global field sensitivity threshold of the entire nociceptive and sensory system has been lowered — every sensory input now has access to pathways it should not normally reach.

The treatment failure profile of fibromyalgia also becomes explicable. Opioids change the chemistry of individual synapses in the ascending pain pathway. They reduce the probability of synaptic transmission at each link in the chain. But they do not restore the geometry of the field. The signals are still distorted, still arriving at wrong addresses, still failing to match the cortical map. Reducing the amplitude of a mis-routed signal is not the same as correctly routing it. Many fibromyalgia patients find that opioids reduce their pain somewhat for a short period before losing effect — which is consistent with chemically damping a geometrically disordered signal while doing nothing to correct the disorder. SSRIs and SNRIs have some benefit in fibromyalgia, and the geometry framework explains why: they alter descending inhibitory tone from the PAG-RVM system, which is the right direction — they are partially restoring the descending field geometry. But they are a blunt instrument applied to a precise geometry problem, which explains why the benefit is real but modest. Pregabalin and gabapentin reduce abnormal field oscillations by blocking voltage-gated calcium channels at synaptic terminals — this is a partially correct geometry approach, reducing the hyperexcitability that distorts field propagation, though without

restoring the structural geometry of axons or cortical maps. Aerobic exercise, consistently among the most effective long-term interventions for fibromyalgia, imposes structured, patterned afferent input from muscle spindles, joint mechanoreceptors, and cutaneous receptors — it feeds the somatosensory system a coherent, organized signal that gradually recalibrates the global map geometry. This is geometry restoration through repetition, and it is why exercise works even when it is initially painful, and why the benefit builds over weeks rather than hours.

7. Diagnostic Blind Spots — Why Medicine Keeps Missing This

Standard pain evaluation looks for tissue damage, inflammation, and neurochemical abnormality. X-rays, MRI, CRP, ESR, CBC, and metabolic panels are the front-line tools. All of them are chemical or structural questions asked of a geometry problem. A nerve that is conducting abnormally because its nodal spacing has been subtly disrupted by chronic inflammation will show a normal MRI. A dorsal horn segment running at twice its normal gain because central sensitization has altered its local field potential will show a normal lumbar MRI. A thalamus broadcasting abnormal oscillatory patterns that misroute somatosensory input will show no signal abnormality on structural neuroimaging. Structure-based imaging cannot see function. And function-based geometry cannot be read from structure alone.

Electromyography and nerve conduction studies are better tools for this question. They measure conduction velocity and amplitude in peripheral nerves — both of which are proxies for the geometry of signal propagation. A nerve with damaged myelin has slowed conduction velocity. A nerve with axon loss has reduced compound action potential amplitude. These are geometric measurements, even if they are not labeled as such. But EMG and NCS have significant limitations in the context of fibromyalgia and many chronic pain conditions. They sample only large myelinated fibers in peripheral nerves. They cannot access the central pathways. They cannot measure the small unmyelinated C-fibers

that carry much of the chronic pain signal. A patient with profound C-fiber small fiber neuropathy can have a completely normal EMG. The test is looking in the wrong place with the wrong instrument.

Quantitative Sensory Testing is the most geometry-aware tool in the standard pain medicine toolkit. QST maps the spatial and threshold boundaries of abnormal sensitivity across the body surface — it measures where signals are being amplified, where touch is being misread as pain, where heat thresholds are shifted, where temporal summation is abnormal. It is mapping the output of the geometry problem even if it does not directly image the geometry itself. QST is underused in fibromyalgia evaluation, underreimbursed, and understandable only to clinicians who already think in terms of central processing rather than peripheral pathology. Its neglect is a genuine clinical failure.

Skin punch biopsy for intraepidermal nerve fiber density is a direct structural proxy for peripheral EMF geometry. It counts the fine-caliber fibers that constitute the peripheral antenna for the nociceptive system. It can directly detect small fiber neuropathy in patients whose standard EMG is normal. Its underuse in fibromyalgia evaluation is a clinical gap of the first order. If 40 to 60 percent of fibromyalgia patients have small fiber neuropathy, then routine skin punch biopsy as part of fibromyalgia workup would identify a substantial subset of patients with a specific, measurable, structural peripheral geometry abnormality — which has direct implications for treatment targeting. The test is available, validated, and not being used systematically. This is a failure of conceptual framing, not a failure of technology.

Electroencephalography and magnetoencephalography can directly image field dynamics at the cortical level. MEG, in particular, captures the magnetic component of neural EMF with millisecond temporal resolution and sub-centimeter spatial resolution. It can directly image the thalamocortical oscillatory patterns that this paper argues are central to fibromyalgia pathophysiology. MEG is a research tool in most institutions, expensive, and not part of any standard pain evaluation protocol. But the fact that it could directly image the primary geometry failure in fibromyalgia — the aberrant thalamocortical oscillations,

the disrupted alpha-band gating, the abnormal spatial patterns of cortical activation — makes its absence from clinical pain evaluation more than a resource limitation. It represents a failure to ask the right question with the right instrument.

Resting-state fMRI connectivity studies have already found in fibromyalgia exactly what the geometry framework predicts — disrupted thalamocortical connectivity, abnormal default mode network activity, altered functional coupling between somatosensory and pain-processing regions. Gracely et al.'s 2002 landmark study demonstrated that fibromyalgia patients show augmented cortical and subcortical activation in response to pressure stimuli that produce no significant activation in controls — the same pressure produces objectively measurable extra cortical activity. This is direct imaging evidence of a system that is amplifying signals it should be filtering. The data are there. But they are being interpreted through a connectivity-map lens — "these regions are abnormally connected" — rather than a field-geometry lens: "the oscillatory architecture that routes signals through these regions is distorted." The gap is not in the data. It is in the conceptual frame used to read the data.

8. Restoration of EMF Geometry — Therapeutic Implications

Chemistry is one lever for modifying field geometry. It is not the only lever, and in many chronic pain and fibromyalgia cases, it is not the most powerful lever. You cannot chemistry your way out of a geometry problem. If the structural geometry of a nerve pathway has been altered, if cortical maps have reorganized around a distorted input pattern, if thalamocortical oscillations are chronically dysrhythmic — chemistry can modulate these processes but it cannot reconstruct them. Reconstruction requires physical, structural, and electrophysiological interventions.

Repetitive, patterned sensorimotor input is the most evidence-grounded geometry restoration strategy available without specialized equipment. Graded motor imagery, mirror therapy, and structured movement programs all work by imposing organized

afferent field patterns on the somatosensory system. In graded motor imagery, the patient progresses through visualizing limb movement, mirror-viewing of the intact limb, and finally actual movement — each step feeding a progressively richer and more geometrically coherent afferent signal into the cortical map of the affected body region. The cortical map, which has reorganized around distorted input, is being retrained by repetitive exposure to correctly shaped afferent fields. The evidence base for this approach in Complex Regional Pain Syndrome and phantom limb pain is solid, and emerging evidence supports its utility in fibromyalgia.

Transcranial Magnetic Stimulation and transcranial Direct Current Stimulation directly impose external field geometry on cortical tissue. rTMS delivers focused magnetic field pulses that induce electrical currents in cortical tissue, modifying local field potentials and altering the excitability of targeted cortical and thalamocortical circuits. Repetitive TMS over the primary motor cortex has FDA clearance for major depression — a condition that shares significant thalamocortical dysrhythmia features with fibromyalgia — and is in active trials for fibromyalgia and chronic pain. The mechanism in the geometry framework is direct: rTMS is applying an external field perturbation to a cortical region whose local field potential is pathologically organized, with the goal of shifting that organization toward a more normal oscillatory geometry. It is not changing a synaptic chemical. It is rewriting the rhythm of the field.

TENS and spinal cord stimulation occupy different anatomical levels of the same intervention principle: external re-imposition of the large-fiber inhibitory field that closes the dorsal horn gate. TENS does this at the peripheral nerve level. Spinal cord stimulation places electrodes on the dorsal epidural surface and delivers continuous electrical field stimulation to the dorsal columns, activating large-fiber afferents at the spinal level and imposing the geometric filter that chronic pain has erased. Spinal cord stimulation is highly effective for a range of neuropathic pain conditions — not equally effective for all, partly because not all neuropathic pain has the same geometry failure. The patients who respond best are those whose primary geometry failure is at the dorsal horn level rather than

cortically — their gate is broken but their cortical maps are still organized. Restoring the gate restores the system.

Sleep restoration is not a secondary intervention in fibromyalgia. It is primary. If slow-wave sleep is disrupted, the nightly thalamocortical field geometry reset does not occur. Every other intervention — exercise, TMS, sensorimotor training — is operating on a system that is being re-disrupted every night. Sleep architecture restoration — whether through cognitive behavioral therapy for insomnia, sodium oxybate (which specifically promotes slow-wave sleep and has shown efficacy in fibromyalgia trials), or careful attention to sleep hygiene and circadian entrainment — should precede or run concurrently with all other pain-specific interventions. A geometry restoration program run in a system that cannot reset overnight is fighting against the clock, literally.

Low-dose naltrexone is an intervention whose mechanism fits the geometry framework precisely, even though it was not designed with that framework in mind. At doses far below its opioid-blocking range (typically 1.5 to 4.5 mg daily), naltrexone antagonizes Toll-like receptor 4 on microglia and astrocytes, reducing glial inflammatory activation. In the geometry framework, this targets the dielectric distortion component of fibromyalgia: it reduces the neuroinflammatory state that changes tissue impedance and distorts local field propagation. Multiple small trials have shown modest but consistent benefit in fibromyalgia, with a favorable side-effect profile. The benefit is modest because glial activation is one of several geometry failure mechanisms — correcting it alone does not restore thalamocortical oscillations or somatosensory maps. But it removes one source of ongoing geometry corruption.

Aerobic exercise is the highest-grade evidence intervention for fibromyalgia that exists, and the geometry framework explains why with precision that the chemical model cannot. Rhythmic aerobic exercise generates massive, highly structured afferent input from muscle spindles, Golgi tendon organs, joint mechanoreceptors, and cutaneous receptors — the entire sensorimotor periphery fires in organized, temporally patterned bursts synchronized with movement. This is a coherent, structured electromagnetic field being fed into the

somatosensory system at high repetition. Over weeks and months of consistent exercise, this patterned input recalibrates the somatosensory map — the cortex learns the geometry of a correctly signaling, active body and reorganizes its maps accordingly. Exercise also drives brain-derived neurotrophic factor (BDNF) production, which supports myelin maintenance and node-of-Ranvier integrity — directly maintaining the peripheral geometry of the EMF propagation system. BDNF also supports the inhibitory interneuron populations in the dorsal horn whose loss opens the gate. Exercise is a full-system geometry restoration intervention, which is why it works when single-target pharmacological interventions do not.

Ketamine infusions at subanesthetic doses block NMDA receptors in the spinal dorsal horn, interrupting the windup and central sensitization loop. In the geometry framework, ketamine is performing a hard reset of the temporal geometry of spinal nociceptive processing. Windup is a temporal geometry pattern — a self-reinforcing cycle of escalating NMDA receptor activation driven by repeated C-fiber input at specific intervals. Blocking NMDA receptors breaks the cycle. The temporal geometry of the nociceptive synapse resets to its baseline. Many fibromyalgia and CRPS patients report significant relief from ketamine infusions that may last weeks to months — far beyond the pharmacological presence of the drug. This is consistent with the interpretation that ketamine is correcting a persistent geometry pattern, not just blocking a chemical. Once the pattern is interrupted, the system may remain in the more normal geometry state for some time before drift back toward pathological windup begins.

The future direction that most logically follows from this framework is closed-loop neurostimulation guided by real-time field geometry measurement. Deep brain stimulation and responsive neurostimulation (RNS) systems already do this in epilepsy: they read the electrographic field of the brain, detect pathological oscillatory patterns, and deliver corrective stimulation to interrupt them. The same principle applied to fibromyalgia would involve continuous monitoring of thalamocortical oscillatory geometry via implanted or non-invasive electrophysiological sensors, with responsive stimulation of appropriate

targets — thalamic relay nuclei, PAG, or primary sensorimotor cortex — delivered when the system drifts toward the pathological oscillatory geometry that drives pain. This is not speculative physics. The technology base exists. The limiting factor is the conceptual frame — closed-loop geometry correction requires that the field be the primary target, which requires accepting the framework this paper is advancing.

9. Conclusion — A New Frame for Old Pain

The chemical model of pain medicine has produced antidepressants, anticonvulsants, opioids, muscle relaxants, NSAIDs, and a growing roster of targeted biologic agents. Each of these helps some patients some of the time. None of them reliably restores function in chronic pain and fibromyalgia. The opioid crisis is, in part, a consequence of applying the wrong model — treating a geometry problem with a chemical that temporarily reduces the subjective magnitude of a mis-routed signal without ever correcting the routing. As the chemical wears off, the geometry is still wrong, the signal is still mis-routed, the pain is still there. The dose escalates. The geometry does not improve.

The geometry model does not discard chemistry. Chemistry is one lever for modifying geometry. GABAergic drugs change the inhibitory architecture of dorsal horn circuits — a geometry change. Sodium channel blockers change the action potential threshold of nociceptive axons — a geometry change in the sense that they alter the conditions under which field propagation initiates. BDNF-promoting interventions support myelin health — a direct geometry maintenance effect. Chemistry and geometry are not competing frameworks. Chemistry is downstream of geometry. The physical structure and dynamic electromagnetic behavior of neural tissue is the primary variable. Chemistry is what you use to nudge it. But nudging is not the same as rebuilding, and rebuilding is what chronic pain and fibromyalgia require.

The neuron is a field-propagation structure. This is not a metaphor and it is not a hypothesis — it is a consequence of electromagnetism applied to a biological system that

generates charge separation. The nervous system is a three-dimensional electromagnetic architecture, organized in space, dynamic in time, and geometrically specific in its encoding of sensory information. Pain is an interpretation of the geometry of arriving signals. It is not a chemical in the blood, not an enzyme activity, not a receptor occupancy. It is what happens when fields propagating through the nervous system arrive at cortical and subcortical structures in patterns that those structures interpret as signaling tissue damage. Chronic pain is what happens when the geometry of those arriving patterns is wrong. Fibromyalgia is what happens when the geometry is wrong everywhere, at every level, simultaneously.

These claims are testable. MEG can directly image the thalamocortical oscillatory patterns this paper identifies as central to fibromyalgia. EEG source modeling can map the spatial geometry of cortical activation patterns during experimental pain stimulation in chronic pain patients versus controls. QST mapping can characterize the spatial geometry of sensitization across the body surface and track its change with treatment. Skin punch biopsy can quantify the peripheral field-propagation architecture in fibromyalgia cohorts and correlate it with central measures. Closed-loop stimulation trials can test whether correcting the oscillatory geometry of thalamocortical circuits reduces pain — and the time course of benefit after stimulation ends will tell us whether the correction is temporary chemical dampening or genuine geometry restoration.

The next decade of pain medicine should be organized around field geometry as the primary target. Not because the geometry model is complete — it is not. Not because it answers every question about chronic pain — it does not. But because it answers questions that the chemical model cannot touch, it explains phenomena that the chemical model cannot accommodate, and it points toward interventions that the chemical model cannot generate. The chemical model asks: what molecule is abnormal? The geometry model asks: what is the shape of the signal doing, and why is it doing that? These questions have produced different experiments, different instruments, and different therapeutic strategies

for a century. One set of questions has produced the opioid crisis. The other set might produce something better.

The map is broken, not the territory. The body is not generating pain because it is damaged in every location the patient reports. The nervous system is generating pain because it has lost the ability to correctly map the signals it is receiving. Fix the map. Restore the geometry. The pain will follow.

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